

MAKIM: Multi-Agent Kernel Integrity Monitor

A multi-AI agent framework for detecting rootkit-like behavior on Linux

Kernel Systems Security Course — Term Project Assignment 1

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Abstract

This report proposes MAKIM (Multi-Agent Kernel Integrity Monitor), a Python user-space system that coordinates specialized AI agents to detect rootkit-like behavior on Linux. Traditional tools rely on static signatures and fail against adaptive threats. MAKIM uses a Scanner Agent to read `/proc` and kernel data, an Anomaly Detector to compare against a trusted baseline, a Rootkit Pattern Agent to flag indicators like hidden PIDs and suspicious modules, and an LLM Analyst Agent (Claude API) for contextual reasoning and remediation advice.

1. Motivation — Why This Functionality Is Needed

The Linux kernel's integrity is the foundation of AI system trustworthiness. A rootkit operating at Ring 0 can silently exfiltrate model weights, manipulate inference results, and intercept multi-agent communications — all invisible to user-space tools. Existing tools (rkhunter, chkrootkit, Tripwire) depend on known-signature databases and produce noisy logs without intelligent triage. They fail against zero-day rootkits and environment-aware attackers.

As Physical AI systems (robots, autonomous vehicles) and AI inference servers proliferate on Linux, this gap becomes a critical vulnerability. The OS needs a security layer that is itself intelligent — capable of reasoning about context, not just matching patterns. Just as DNS was added to BSD when the internet emerged, the AI era demands a new kernel security primitive: a multi-agent monitor that can detect, explain, and respond to kernel-level threats.

2. Background

Rootkits achieve stealth through Direct Kernel Object Manipulation (DKOM) — removing a process from the kernel task list — and syscall table hooking, which intercepts calls like `getdents64` to hide files. The Linux `/proc` filesystem exposes kernel state (module list, process maps, memory layout) readable from user-space, making it an ideal data source for a user-space monitor.

Multi-agent AI systems coordinate specialized agents under an orchestrator. Protocols like MCP (Model Context Protocol, Anthropic 2024) and A2A (Agent-to-Agent, Google 2025) standardize agent-tool communication. LLM-powered agents add reasoning capabilities that rule-based systems cannot replicate.

3. Proposed System: MAKIM

MAKIM runs as a Python daemon on Ubuntu Linux (kernel 6.8+), requiring no kernel modules. It coordinates five specialized agents:

Agent	Responsibility
Scanner Agent	Reads /proc/modules, /proc/[pid]/maps, dmesg, lsmod
Anomaly Detector	Differs current snapshot vs. trusted baseline; flags new modules, hidden PIDs (DKOM indicator)
Rootkit Pattern Agent	Checks suspicious module names, anon exec mappings, dmesg taint warnings, C2 ports
LLM Analyst Agent	Sends structured findings to Claude API; returns threat level + remediation steps
Report Agent	Produces terminal output + JSON log; triggers alert if severity >= HIGH

References & AI Tool Usage

AI tools: Claude (claude-sonnet-4-20250514) for brainstorming and writing assistance.